KidsRoute
School Transport Vehicle Management System

CO2060 – Software Systems Design Project

Team AlphaWolves (48)

Real-world Problem Statement

KidsRoute

- **Lack of Transparency**
  Parents cannot track vehicle location

- **Uncertain Timing**
  Long waiting times at pickup points

- **Manual Management**
  Drivers handle attendance and routes manually

Target Users & Solution Goal

KidsRoute

- **Parents**
  Track child's journey
  Receive notifications

- **Drivers**
  Manage routes and attendance

- **Admins**
  Monitor vehicles and operations

- **Goals:**
  Improve safety
  Increase transparency
  Simplify transport management

Progress Since Proposal

- Role-Based Authentication (Admin, Driver, Parent)
- Vehicle & Route Management
- User Dashboards & Profiles
- Journey Tracking APIs (start/end journeys)
- Student Pickup/Drop Logging
- Notification System (basic)

Challenges faced & Solutions

Challenge	Solution
Real-time tracking complexity	Implemented API-based tracking (polling)
Role-based access control	Used JWT authentication
Data consistency across modules	Designed relational schema in PostgreSQL
Integration between frontend & backend	Modular API structure

Team Collaboration & Contribution

Task Distribution

- K.A.H.G.D.Sandeepa (E/22/354) — authentication & core system
- K.I.Sewmini (E/22/272) — driver, vehicle & route management
- H.D.Lokugamage (E/22/214) — tracking & notifications
- S.I.Gunawardhana (E/22/127) — admin dashboard & integration

GitHub Commit History Link:

https://github.com/cepdnaclk/e22-co2060-School-Van-Management-System/commits/main/

Demonstration of the MVP

User Feedback & Validation

Impact of Feedback on Development

Semester 4 Development Plan

Task \ Week	1	2	3	4	5	6	7	8	9	10	11	12	13
Google Map Integration													
Absent Feature													
Emergency Alerts													
Admin Analytics Dashboard													
UI Improvement													
Testing & Bug Fixing													
Final Deployment & Documentation													

Planned Feature Enhancement

- **High Priority**
  Real-time map tracking (Google Maps + WebSockets)

- **Smart Features**
  Mark student as absent

- **Safety Features**
  Emergency SMS alerts

- **Admin Improvements**
  Analytics dashboard

Testing Strategy

- **Unit Testing**
  Test individual APIs and components

- **Integration Testing**
  Test frontend + backend communication

- **Acceptance Testing**
  Validate with real users (parents/drivers)

Deployment & Documentation

- **Deployment**
  Frontend - Netlify
  Backend - Render
  Database - PostgreSQL (Supabase)

- **Documentation**
  API documentation
  User guide (parents & drivers)
  Developer documentation

GitHub Repository

https://github.com/cepdnaclk/e22-co2060-School-Van-Management-System

CN ePortfolio

Thank You!

Team AlphaWolves